



1. The average age of a group going for picnic is 16 yrs 20 persons with average age 15 yrs join them so that the average age becomes 15.5 years. Then the number of persons who were initially going for picnic: **(Average) NIMCET-2007**

(a)10 (b)20 (c)18 (d)none

2. A train travels at an average speed of 100 kmph stopping for 3 minutes after every 75 kms. How much time it will take to reach a station which is 600 km from the point of start : **(Average) NIMCET-2007**

(a)6 hrs. 24 min. (b)6 hrs. 21 min.
(c)6 hrs. (d)none

3. If $A_1 = \{3\}$, $A_2 = \{5, 7, 9\}$, $A_3 = \{11, 13, 15, 17, 19\}$, $A_4 = \{21, 23, 25, 27, 29, 31, 33\}$ and so on, what is the average of the numbers of the set A_{20} ? **(Average)**

(a)761 (b)763 (c)765 (d)767

4. The average age of husband and wife was 22 years when they were married five years back. What is the present average age of the family if they have a three year old child? **(Average) NIMCET-2011**

(a)19 Years (b)25 Years
(c)27 Years (d) $28\frac{1}{2}$ Years

5. Average age of students of an adult school is 40 yr. 120 new students whose average age is 32 yr joined the school. As a result the average age is decreased by 4 yr. The number of students of the school after joining of the new student is **(Average) NIMCET-2012**

(a) 1200
(b) 120
(c) 360
(d) 240

6. The arithmetic mean of 2^{10} and 2^{20} is
(a) 2^{15} (b) $2^5 + 2^{10}$ (c) $2^9 + 2^{20}$ (d) $2^9 + 2^{19}$

(Average) NIMCET-2014

7. A set consecutive positive integers beginning with 1 is written on the blackboard. A student came along and erased one number. The average of the remaining numbers is 35717. What was the number erased?

(Average) NIMCET-2020

(a)7 (b)8 (c)9 (d)None

8. Sum of ages of Anu and bhanu is 10 yr more than sum of ages of Bhanu, Chanu and Dhanu. Average age of Chanu and dhanu is 19 yrs. Find the average age of Anu and Dhanu if Dhanu is 10 years elder than Chanu. **(Average) NIMCET-2020**

(a)36 yr (b)30 yr (c)25 yr (d)31 yr

9. The mean of consecutive positive integers from 2 to n is

(Average) NIMCET-2024

(a) $\frac{n+2}{2}$ (b) $\frac{n(n+1)}{2}$
(c) $\frac{n+1}{2}$ (d) $\frac{n-1}{2}$

ANSWER

1	b	2	b	3	b	4	a	5	d
6	d	7	a	8	a	9	a		

1. (b)

Let Initial person = n

$$\bar{x}_n = 16 \text{ years}$$

Another person = 20

$$\bar{x}_m = 15 \text{ years}$$

Average = 15.5 year

$$\frac{\bar{x}_n n + \bar{x}_m m}{n+m} = 15.5$$

$$\frac{16 \times n + 15 \times 20}{20+n} = 15.5$$

$$(0.5 n) = (20 \times 0.5)$$

$$n = 20$$

2. (b)

Average speed = 100 km/h

with this speed train cover, 600 km in 6 hr but train stop for 3 min after 75km

$$= \frac{600}{75} = 8 \text{ time}$$

But in 8th time train will be at destiny to train stop 7 time

Time take = 6 hr + 7 × 3 min

6 hr 21 min option (b)

3. (b)

In A_{20} , there will be $2 \times 20 - 1 = 39$ numbers in AP with common difference 2.

Also, the first term of A_{20} will be

$$3 + 2 \times 3 + 2 \times 5 + 2 \times 7 + \dots + 2 \times 37 + 2$$

$$= 3 + 2(1 + 3 + 5 + 7 + \dots + 37)$$

$$= 3 + 2 \times \frac{19}{2} [2 + 18 \times 2] = 3 + 2 \times 19 \times 19 = 725$$

Average of 39 numbers of A_{20}

$$= \frac{39}{2} [2 \times 725 + (39 - 1) \times 2] = 725 + 38 = 763$$

4.

(a) Five years ago total age of husband and wife

$$22 \times 2 = 44 \text{ yr}$$

Present age of husband and wife = $44 + 2 \times 5$

$$44 + 10 = 54 \text{ yr}$$

$\therefore$ Total age of the family = $54 + 3 = 57 \text{ yr}$

Hence, average age of the family = $\frac{57}{3} = 19 \text{ yr}$

5.

(d) Let the total number of students before joining new students = x.

$\Rightarrow$ After joining new 120 students = $x + 120$

Now, by condition,

$$x \times 40 + 120 \times 32 = (x + 120) \times 36$$

$$\Rightarrow 40 \times 3840 = 36x + 4320$$

$$\Rightarrow 4x = 480$$

$$\therefore x = 120$$





∴ Total number of students = $x + 120 = 120 + 120 = 240$

6. (d) Arithmetic mean of 2^{10} and $2^{20} = \frac{2^{10} + 2^{20}}{2}$
 $= \frac{2^9 + 2^{19}}{2} = 2^9 + 2^{19}$

7. (a) Since all the values are integers, their sum must be an integer.
Sum = Number $\times$ Average

Number of integers after erasing must be a multiple of 17.

As the average of n consecutive numbers is $\left(\frac{n+1}{2}\right)$, then number of integers must be around 69.

After erasing the number of integers is 68 (as 68 is a multiple of 17) and before erasing, the number of integers is 69.

Let the erased number be x .

$$\frac{69 \times 35 - x}{68} = \frac{602}{17}$$

$$x = 7$$

8. (a) Sum of ages of Anu and Bhanu is 10 years more than sum of ages of Bhanu, Chanu and Dhanu,

$$\text{Anu} + \text{Bhanu} = (\text{Bhanu} + \text{Chanu} + \text{Dhanu}) + 10$$

$$\text{Chanu} + \text{Dhanu} + 10 = \text{Anu} \quad \dots\dots\dots(i)$$

Average age of Chanu and Dhanu is 19 years

$$\frac{(\text{Chanu} + \text{Dhanu})}{2} = 19$$

$$\text{Chanu} + \text{Dhanu} = 38 \text{ year} \quad \dots\dots\dots(ii)$$

Putting the value of (ii) in (i) we get,

$$38 + 10 = \text{Anu}$$

So, age of Anu is 48 years

Dhanu is 10 years elder than Chanu

$$\text{Dhanu} - 10 = \text{Chanu}$$

Putting the value of (iii) in (ii), we get,

$$\text{Dhanu} - 10 + \text{Dhanu} = 38$$

$$2 \text{ Dhanu} = 38 + 10$$

$$2 \text{ Dhanu} = 48$$

$$\text{Dhanu} = 24 \text{ yrs.}$$

Average age of Anu and Dhanu $(48 + 24)$

$$= \frac{72}{2} = 36 \text{ years}$$

∴ The answer is 36 yr.

∴ The correct answer is option (a).

9. (a)
Average
1,2,(3),4,5

$$\frac{l+F}{2} = \frac{5+1}{2} = \frac{n+2}{2}$$